

① data mining

data mining is the process of discovering useful patterns, relationship and knowledge from large data sets using statistical.

② two data mining tasks

* classification * clustering

steps in KDD

* data cleaning → remove noise and missing values
* data integration → combine data from multiple.

Role of Data Preprocessing :-

- * improves data quality
- * remove inconsistencies and redundancy.
- * increase mining accuracy.

② data cleaning :-

data cleaning is the process of detecting and correcting errors, missing values, duplicate

two data quality issues :-

* missing values * noisy data.

Handling missing values

* ignore the record

example :-

Age values 20, 21, NULL, 23 → Replace NULL with mean ($21.3 \approx 21$)

②
a

data integration

It is the process of combining data from multiple source into a single, consistent dataset.

challenges:-

- * data redundancy
- * data inconsistency

Handling redundancy

- * remove duplicate records
- * correlation analysis

Handling inconsistency:-

- * standardize formats
- * resolve conflicting values.

②
b

covariance analysis

covariance measures how two variables change together.

population covariance:-

$$\text{cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

sample covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

3
a

covariance interpretation

Given

$$A = (2, 3, 5, 4, 6)$$

$$B = (5, 8, 10, 11, 14)$$

$$\text{mean}(A) = 4$$

$$\text{mean}(B) = 9.6$$

products:-

$$(-2)(-4.6) = 9.2$$

$$(-1)(-1.6) = 1.6$$

$$(1)(0.4) = 0.4$$

$$(0)(1.4) = 0$$

$$2(4.4) = 8.8$$

$$\boxed{\text{Total} = 20}$$

$$\text{sample covariance} = \frac{20}{4} = 5$$

covariance = 5 (positive)

* covariance is positive, the stocks are not independent.

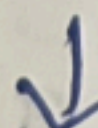
b

Architecture of Data Mining system

data source



data warehouse



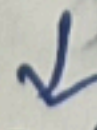
preprocessing



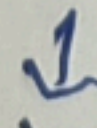
data mining engine



pattern evaluation



knowledge base



user interface

Find covariance

Given $x = \{4, 5, 6, 7, 9\}$ $y = \{8, 3, 7, 5, 6\}$

$$\text{mean}(x) = 6.2$$

$$\text{mean}(y) = 5.8$$

Products:

$$(-2.2)(2.2) = -4.84$$

$$(0.8)(-0.8) = -0.64$$

$$(-1.2)(-2.8) = +3.36$$

$$(2.8)(0.2) = 0.56$$

$$(-0.2)(1.2) = -0.24$$

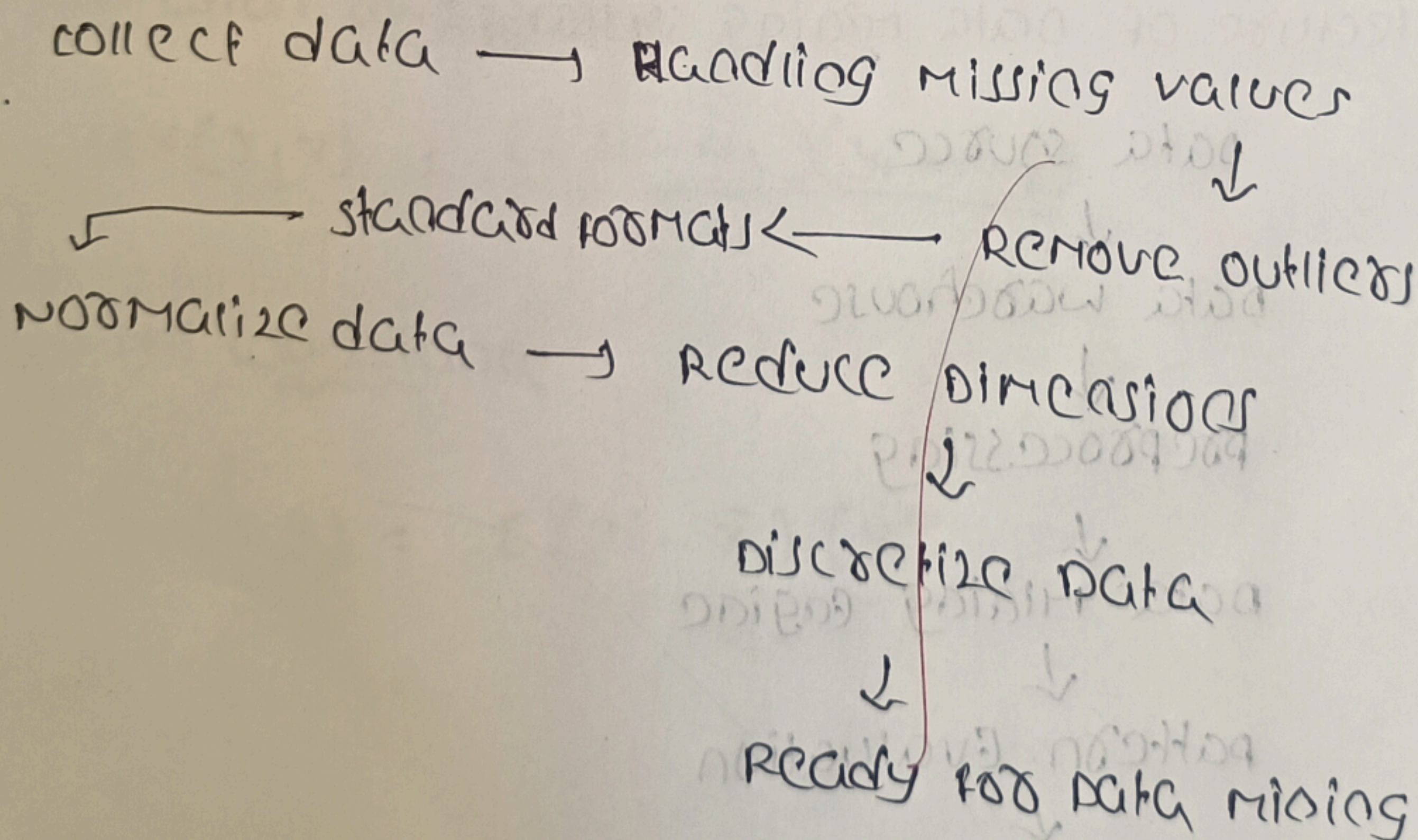
$$\boxed{\text{SUM} = -1.8}$$

$$\text{sample covariance} = \frac{-1.8}{4} = -0.45$$

major steps in data preprocessing

- * Data cleaning
- * Data integration
- * Data transformation
- * Data reduction
- * Data discretization

Pipeline



⑤
④
Given data (30 values)

Bin size = $\frac{30}{3} \Rightarrow 10$ values per bin

Bin 1:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22. Mean = 18.4

~~Bin 1~~ :- smoothed Bin :-

18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4,

Bin 2:-

25, 25, 25, 25, 30, 33, 33, 35, 35, 35. Mean = 30.1

smoothed :- 30.1 repeat 10 times

Bin 3:- 35, 36, 40, 45, 46, 52, 70. Mean = 46.3

Replace each by 46.3

⑥ Data Discretization:-

Data discretization converts continuous numerical values into intervals.

Importance:-

- * Reduce data complexity
- * Remove noise.

equal width Binning

- * All bins have equal range
- * Simple and fast

equal frequency

- * Handles skewed data better
- * Bin widths may vary.